

Normal Hilbert semigroups and the fractional-range stability condition

Run fa_banach_001, arXiv:math/0512460, partial answer

Source problem

Borichev, Chill, and Tomilov prove in [1] that, for a bounded C_0 -semigroup generated by A on a Banach space with Fourier type $p \in (1, 2]$, density of the set

$$M_p(A) := \left\{ x : \lim_{\alpha \rightarrow 0+} \alpha^{(p-1)/p} R(\alpha + i\beta, A)x = 0 \text{ for every } \beta \in \mathbb{R} \right\}$$

implies strong stability. They also prove the algebraic sufficient condition

$$\bigcap_{\beta \in \mathbb{R}} \text{Rg}(i\beta - A)^{1/p} \text{ is dense.}$$

Remark 5.11 asks whether these conditions characterize stability in the Hilbert case $p = 2$.

Remark 5.11. Corollary 5.10 improves the stability result from [9] Corollary 4.6]; the exponent $\frac{1}{p}$ in (5.5) was there replaced by an exponent $\gamma > \frac{1}{p}$.

Corollary 5.10 is optimal in the following sense: for every $p \in [1, 2]$ there exist a Banach space X having Fourier type p and a *nonstable* bounded semigroup with generator A such that for every $\gamma \in (0, \frac{1}{p})$

$$\bigcap_{\beta \in \mathbb{R}} \text{Rg}(i\beta - A)^\gamma \text{ is dense in } X;$$

see [8] Example 4.9].

Note that the stability condition from Corollary 5.10 is sufficient but in general not necessary for stability, except possibly in the case $p = 2$, [8] Example 4.1]. In fact, it is an open problem whether the condition (5.5) (or the stability condition in Theorem 5.7) characterizes stability in Banach spaces with Fourier type $p = 2$, i.e in Hilbert spaces.

Proof of Corollary 5.10 The case $p = 1$ has been proved in [4] Theorem 2.4]. The assumption implies that $\text{Rg}(i\beta - A)$ is dense in X . By the fractional mean ergodic theorem, [19] Proposition 2.3], [29] Proposition 2.2], for every $x \in X$ and every $p > 1$

Partial answer

Theorem 1. *Let A be a normal generator of a bounded C_0 -semigroup $(T(t))_{t \geq 0}$ on a complex Hilbert space H . Then the following are equivalent:*

1. $(T(t))_{t \geq 0}$ is strongly stable;
2. $M_2(A)$ is dense in H ;
3. $\bigcap_{\beta \in \mathbb{R}} \text{Rg}(i\beta - A)^{1/2}$ is dense in H .

Proof. The implications (2) $\Rightarrow$ (1) and (3) $\Rightarrow$ (1) are Theorem 5.7 and Corollary 5.10 of [1], specialized to $p = 2$. It remains to prove the converse under normality.

Let E denote the spectral measure of A . Since A generates a bounded normal semigroup, the spectral support is contained in the closed left half-plane and

$$T(t) = e^{tA} = \int e^{tz} dE(z).$$

Strong stability is equivalent to $E(i\mathbb{R}) = 0$. Indeed, if $E(i\mathbb{R})x = x$, then $\|T(t)x\| = \|x\|$ for all $t \geq 0$, so strong stability forces $x = 0$. Conversely, if $E(i\mathbb{R}) = 0$, then $|e^{tz}| \leq 1$ and $e^{tz} \rightarrow 0$ for E -almost every spectral point, so dominated convergence gives $\|T(t)x\| \rightarrow 0$ for every $x \in H$.

Assume now that the semigroup is strongly stable. For $\varepsilon > 0$, set

$$H_\varepsilon := E\{z : \operatorname{Re} z \leq -\varepsilon\}H.$$

Because $E(i\mathbb{R}) = 0$, the union $\bigcup_{\varepsilon>0} H_\varepsilon$ is dense in H . Fix $x \in H_\varepsilon$ and $\beta \in \mathbb{R}$. On the spectral support of x ,

$$|i\beta - z| \geq \varepsilon.$$

Hence the multiplier $(i\beta - z)^{-1/2}$ is bounded there. By the normal functional calculus,

$$y_\beta := \int (i\beta - z)^{-1/2} dE(z)x$$

belongs to H , and

$$(i\beta - A)^{1/2}y_\beta = x.$$

Thus $x \in \operatorname{Rg}(i\beta - A)^{1/2}$ for every β , and so

$$\bigcup_{\varepsilon>0} H_\varepsilon \subset \bigcap_{\beta \in \mathbb{R}} \operatorname{Rg}(i\beta - A)^{1/2}.$$

The latter intersection is therefore dense.

The same spectral gap proves density of $M_2(A)$. If $x \in H_\varepsilon$, then for every $\beta \in \mathbb{R}$,

$$\left\| \alpha^{1/2} R(\alpha + i\beta, A)x \right\| \leq \frac{\alpha^{1/2}}{\varepsilon} \|x\| \rightarrow 0 \quad (\alpha \rightarrow 0+).$$

Thus $\bigcup_{\varepsilon>0} H_\varepsilon \subset M_2(A)$, and $M_2(A)$ is dense. □

Remark 1. *The normality hypothesis is essential for this packet's proof. The argument uses the spectral theorem to manufacture a common dense subspace lying a positive distance from the entire imaginary axis. It does not address the source problem for arbitrary Hilbert-space generators.*

Ledger note

Cheap run indexes were searched for arXiv:0512460 and the core stability keywords; no duplicate packet was found. A web search on 2026-07-03 for the exact source title and the Hilbert/Fourier-type stability question found the source paper but no later public settlement of the full problem.

References

- [1] A. Borichev, R. Chill, and Y. Tomilov, *Uniqueness theorems for (sub-)harmonic functions with applications to operator theory*, arXiv:math/0512460, 2005.